

Equations with Variables on Both Sides

① $7x = 4 + 3x$ ② $-5y + 3 = 2 + 6y$
 $-3x + 7x = 4 + 3x - 3x$ $5y - 5y + 3 = 2 + 6y + 5y$

$$4x = 4$$

$$\frac{4x}{4} = \frac{4}{4}$$

$$\boxed{x = 1}$$

$$3 = 2 + 11y$$

$$-2 + 3 = 2 + 11y - 2$$

$$1 = 11y$$

$$\frac{1}{11} = \frac{11y}{11}$$

$$\boxed{\frac{1}{11} = y}$$

③ $7 - 3z - 3 = -9z + 6$

$$4 - 3z = -9z + 6$$

$$3z + 4 - 3z = -9z + 6 + 3z$$

$$4 = -6z + 6$$

$$-6 + 4 = -6z + 6 - 6$$

$$-2 = -6z$$

$$\frac{-2}{-6} = \frac{-6z}{-6}$$

$$\boxed{\frac{1}{3} = z}$$

④ $\frac{2x}{5} - 2 = 6 + \frac{3x}{4}$

$$-\frac{3x}{4} + \frac{2x}{5} - 2 = 6 + \frac{3x}{4} - \frac{3x}{4}$$

$$-\frac{15x}{20} + \frac{8x}{20} - 2 = 6$$

$$-\frac{7x}{20} - 2 = 6$$

$$2 - \frac{7x}{20} - 2 = 6 + 2$$

$$-\frac{7x}{20} = 8$$

$$(20)\left(-\frac{7x}{20}\right) = 8(20)$$

$$-7x = 160$$

$$\frac{-7x}{-7} = \frac{160}{-7}$$

$$\boxed{x = \frac{160}{-7}}$$

⑤ $4x - 3 + 7x = 9x + 5$

$$-9x + 11x - 3 = 9x + 5 - 9x$$

$$2x - 3 = 5$$

$$3 + 2x - 3 = 5 + 3$$

$$2x = 8$$

$$\frac{2x}{2} = \frac{8}{2}$$

$$\boxed{x = 4}$$

⑥ $-3y + 5 + 6y = y + 2$

$$3y + 5 = y + 2$$

$$-y + 3y + 5 = y + 2 - y$$

$$-5 + 2y + 5 = 2 - 5$$

$$2y = -3$$

$$\frac{2y}{2} = \frac{-3}{2}$$

$$\boxed{y = -\frac{3}{2}}$$

⑦ $-6z + 4 - 5z = 5z$

$$-11z + 4 = 5z$$

$$11z - 11z + 4 = 5z + 11z$$

$$4 = 16z$$

$$\frac{4}{16} = \frac{16z}{16}$$

$$\boxed{\frac{1}{4} = z}$$

⑧ $5 - \frac{x}{8} = \frac{x}{2} + 1 - \frac{3x}{8}$

$$5 - \frac{x}{8} = \frac{4x}{8} + 1 - \frac{3x}{8}$$

$$5 - \frac{x}{8} = \frac{x}{8} + 1$$

$$\frac{x}{8} + 5 - \frac{x}{8} = \frac{x}{8} + 1 + \frac{x}{8}$$

$$5 = \frac{2x}{8} + 1$$

$$5 = \frac{x}{4} + 1$$

$$-1 + 5 = \frac{x}{4} + 1 - 1$$

$$4 = \frac{x}{4}$$

$$(4)4 = \frac{x}{4}(4)$$

$$\boxed{16 = x}$$

$$\textcircled{9} -1+3y = -y+4+2y$$

$$-1+3y = y+4$$

$$-y-1+3y = y+4-y$$

$$-1+2y = 4$$

$$|-1+2y = 4+|$$

$$2y = 5$$

$$\frac{2y}{2} = \frac{5}{2}$$

$$\boxed{y = 5/2}$$

$$\textcircled{10} -3z+1 = 2-8z$$

$$-3z+1 = 2-8z$$

$$8z-3z+1 = 2-8z+8z$$

$$5z+1 = 2$$

$$-1+5z+1 = 2-1$$

$$5z = 1$$

$$\frac{5z}{5} = \frac{1}{5}$$

$$\boxed{z = 1/5}$$

$$\textcircled{11} 9x-3 = 2x$$

$$-9x+9x-3 = 2x-9x$$

$$-3 = -7x$$

$$\frac{-3}{-7} = \frac{-7x}{-7}$$

$$\boxed{\frac{3}{7} = x}$$

$$\textcircled{12} -8z-4z+8 = -1+3z-6z$$

$$-12z+8 = -1-3z$$

$$12z-12z+8 = -1-3z+12z$$

$$1+8 = -1+9z+1$$

$$9 = 9z$$

$$\frac{9}{9} = \frac{9z}{9}$$

$$\boxed{1 = z}$$

$$\textcircled{13} -1+5-7y+2 = 5y+6-7y$$

$$6-7y = -2y+6$$

$$7y+6-7y = -2y+6+7y$$

$$6 = 5y+6$$

$$-6+6 = 5y+6-6$$

$$0 = 5y$$

$$\frac{0}{5} = \frac{5y}{5}$$

$$\boxed{0 = y}$$

$$\textcircled{15} 5x-8+1 = 7x+3-x$$

$$-5x+5x-7 = 6x+3-5x$$

$$-7 = x+3$$

$$-3-7 = x+3-3$$

$$\boxed{-10 = x}$$

$$\textcircled{14} -9z+3z = z-1+3z$$

$$-6z = 4z-1$$

$$-4z-6z = 4z-1-4z$$

$$-10z = -1$$

$$\frac{-10z}{-10} = \frac{-1}{-10}$$

$$\boxed{z = 1/10}$$

$$\textcircled{16} 9 - \frac{y}{6} - \frac{2y}{3} = 4 + \frac{5y}{6}$$

$$9 - \frac{y}{6} - \frac{4y}{6} = 4 + \frac{5y}{6}$$

$$\frac{5y}{6} + 9 - \frac{5y}{6} = 4 + \frac{5y}{6} + \frac{5y}{6}$$

$$9 = 4 + \frac{10y}{6}$$

$$9 = 4 + \frac{5y}{3}$$

$$-4+9 = 4 + \frac{5y}{3} - 4$$

$$5 = \frac{5y}{3}$$

$$(3) 5 = \frac{5y}{3} (3)$$

$$15 = 5y$$

$$\frac{15}{5} = \frac{5y}{5}$$

$$\boxed{3 = y}$$

$$(17) \quad 12x - 15x + 10 = 3 + 8x + 2x$$

$$-3x + 10 = 3 + 10x$$

$$3x - 3x + 10 = 3 + 10x + 3x$$

$$10 = 3 + 13x$$

$$-3 + 10 = 3 + 13x - 3$$

$$7 = 13x$$

$$\frac{7}{13} = \frac{13x}{13}$$

$$\boxed{\frac{7}{13} = x}$$

$$(19) \quad \frac{3y}{5} + 1 - \frac{3y}{5} = -\frac{y}{10} + 4$$

$$\frac{y}{5} + 1 = -\frac{y}{10} + 4$$

$$\frac{y}{10} + \frac{2y}{5} + 1 = -\frac{y}{10} + 4 + \frac{y}{10}$$

$$\frac{y}{10} + \frac{2y}{10} + 1 = 4$$

$$-1 + \frac{3y}{10} + 1 = 4 - 1$$

$$\frac{3y}{10} = 3$$

$$(10) \quad \frac{3y}{10} = 3(10)$$

$$3y = 30$$

$$\frac{3y}{3} = \frac{30}{3}$$

$$\boxed{y = 10}$$

$$(18) \quad 7 - 5y = -6y - y$$

$$7 - 5y = -7y$$

$$5y + 7 - 5y = -7y + 5y$$

$$7 = -2y$$

$$\frac{7}{-2} = \frac{-2y}{-2}$$

$$\boxed{-\frac{7}{2} = y}$$

$$(20) \quad 7x - 10x + 1 = -2x$$

$$-3x + 1 = -2x$$

$$3x - 3x + 1 = -2x + 3x$$

$$\boxed{1 = x}$$

$$(21) \quad 6 - 2x + 4x = x - 9x$$

$$6 + 2x = -8x$$

$$-2x + 6 + 2x = -8x - 2x$$

$$6 = -10x$$

$$\frac{6}{-10} = \frac{-10x}{-10}$$

$$\boxed{-\frac{3}{5} = x}$$

$$(22) \quad -3y + 2y - 4 = 10y$$

$$-y - 4 = 10y$$

$$y - y - 4 = 10y + y$$

$$-4 = 11y$$

$$\frac{-4}{11} = \frac{11y}{11}$$

$$\boxed{-\frac{4}{11} = y}$$

$$(23) \quad 8x - 4 + 2x = 9x - 4$$

$$-4 + 10x = 9x - 4$$

$$-9x - 4 + 10x = 9x - 4 - 9x$$

$$x - 4 = -4$$

$$4 + x - 4 = -4 + 4$$

$$\boxed{x = 0}$$

$$(24) \quad -2 + 6y = 7y + 3 - 5$$

$$-2 + 6y = 7y - 2$$

$$-6y - 2 + 6y = 7y - 2 - 6y$$

$$-2 = y - 2$$

$$2 - 2 = y - 2 + 2$$

$$\boxed{0 = y}$$

$$(25) 9x - 10x + 1 = 8x + 3$$

$$-x + 1 = 8x + 3$$

$$x - x + 1 = 8x + 3 + x$$

$$1 = 9x + 3$$

$$-3 + 1 = 9x + 3 - 3$$

$$-2 = 9x$$

$$\frac{-2}{9} = \frac{9x}{9}$$

$$\boxed{-\frac{2}{9} = x}$$

$$(26) -y + 4 + 2y = -2 - 3y$$

$$y + 4 = -2 - 3y$$

$$-y + y + 4 = -2 - 3y - y$$

$$4 = -2 - 4y$$

$$2 + 4 = -2 - 4y + 2$$

$$6 = -4y$$

$$\frac{6}{-4} = \frac{-4y}{-4}$$

$$\boxed{-\frac{3}{2} = y}$$

$$(27) 10x + 2 - 11x = 8 - 7x + 1 - 4x$$

$$-x + 2 = -11x + 9$$

$$x - x + 2 = -11x + 9 + x$$

$$2 = -10x + 9$$

$$-9 + 2 = -10x + 9 - 9$$

$$-7 = -10x$$

$$\frac{-7}{-10} = \frac{-10x}{-10}$$

$$\boxed{\frac{7}{10} = x}$$

$$(28) 6y - 8y = 4 + 3y + 5y$$

$$-2y = 4 + 8y$$

$$-8y - 2y = 4 + 8y - 8y$$

$$-10y = 4$$

$$\frac{-10y}{-10} = \frac{4}{-10}$$

$$\boxed{y = -\frac{2}{5}}$$